

1 MOTIVATION

- Multilingual IFT data is scarce, skewed toward English, lacks systematic curation
- Llama-3 IFT dataset: only 3.01% multilingual samples
- No language-agnostic quality scoring for multilingual data
- Data diversity studied only in English settings
- Superficial Alignment Hypothesis (SAH) unverified in multilingual contexts

Three Challenges Addressed:

- No extensible quality scoring for multilingual IFT
 - QSM with triplet loss
- Diversity selection limited to English
 - DAS inspired by MMR
- SAH unverified in multilingual settings
 - 1K → 52K study, 8 langs

2 METHOD: M-DaQ FRAMEWORK

Stage 1: Quality Scoring Model (QSM)

- Fine-tuned on ~2.3K expert-revised samples × 18 languages (from MIDB)
- Triplet loss: instruction → positive (expert-revised) vs negative (original/MT)
- Language-agnostic quality signal from embedding space

Stage 2: Diversity-Aware Selection (DAS)

- MMR-inspired: Quality ≈ Relevance, Diversity ≈ Novelty
- Two-stage pipeline reduces $O(n^2)$ → $O(n \log n)$
 - Stage A: Select top-n by QSM score (quality subset)
 - Stage B: Greedily augment from uncovered clusters (diversity subset)
- Ratio $n_{\text{quality}} : n_{\text{diversity}} = 6:1$ (empirically tuned)

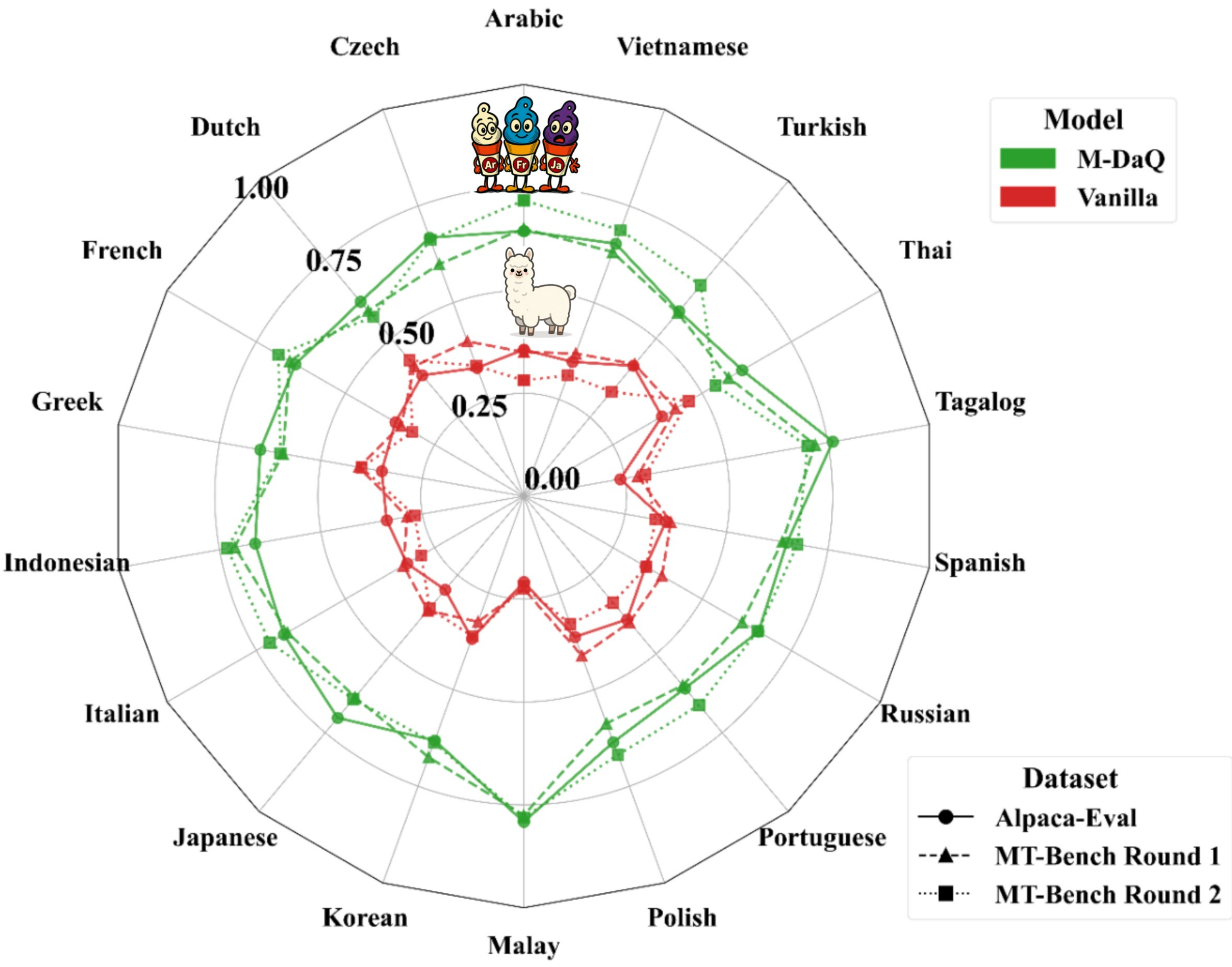
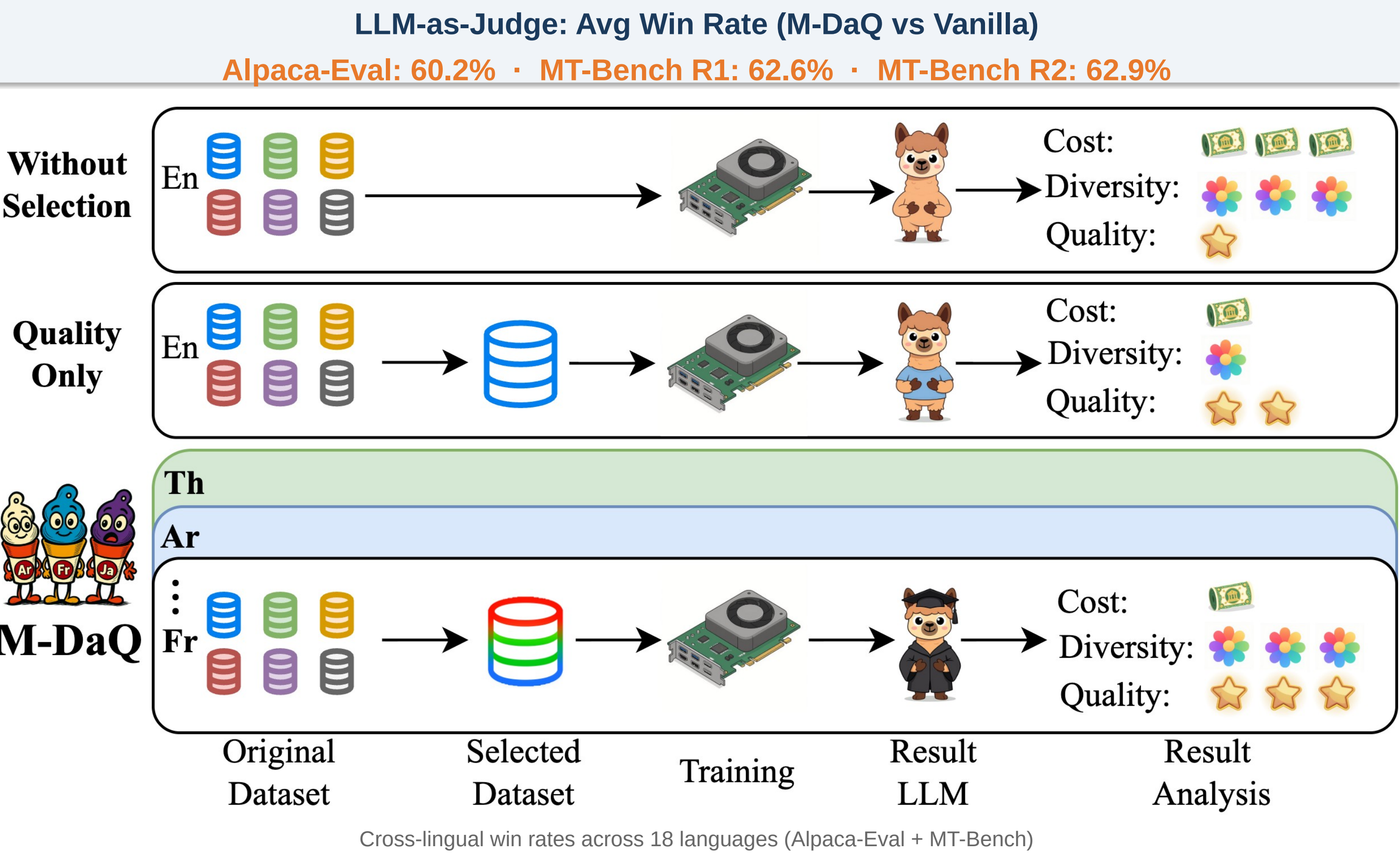


Figure: M-DaQ two-stage pipeline (QSM + DAS)

3 KEY RESULTS



- Consistent improvement across all 18 languages
- Gains MORE pronounced in low-resource languages (Tagalog, Malay)
- M-DaQ mitigates linguistic imbalance and data scarcity

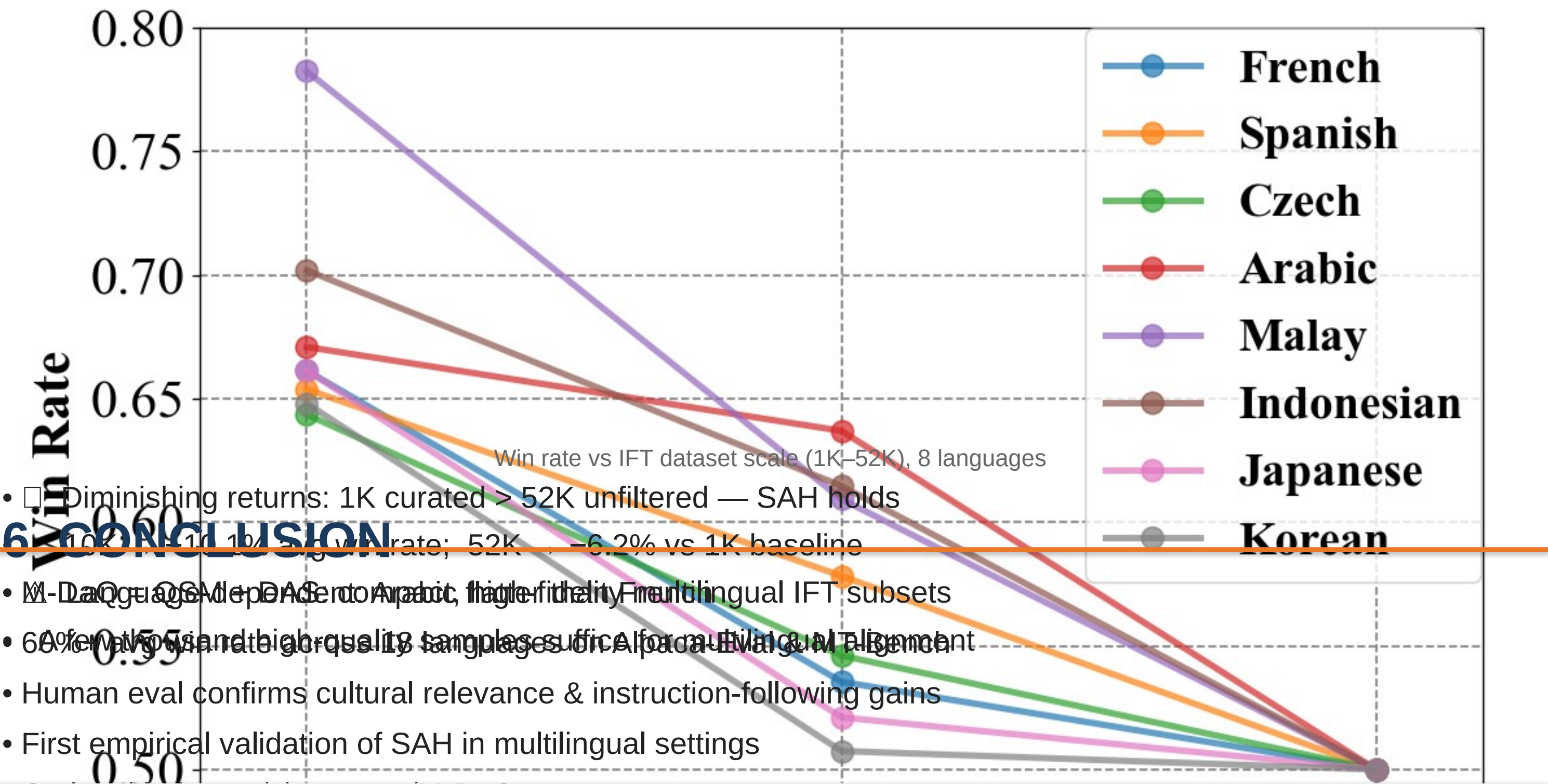
4 HUMAN EVALUATION

6 Languages · 900 Samples · 58 Person-Hours

Language	Alpaca-Eval	MT-Bench R1	MT-Bench R2
Japanese	86%	84%	80%
Korean	94%	94%	94%
Russian	70%	86%	84%
Portuguese	80%	78%	84%
Greek	58%	76%	82%
French	80%	92%	88%
Average	78.0%	85.0%	85.3%

5 SAH IN MULTILINGUAL SETTINGS

First systematic investigation of SAH across languages.



- Diminishing returns: 1K curated > 52K unfiltered — SAH holds
- M-DaQ (curated) significantly outperforms baseline (unfiltered) across all languages, with a notable gain in Malay and Indonesian
- Human eval confirms cultural relevance & instruction-following gains
- First empirical validation of SAH in multilingual settings

Code & Resources
github.com/zhaochunghuang/M-DaQ

Paper
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